PO096

CENTRAL NERVOUS SYSTEM (CNS) VASCULITIS IN ACTIVATED PHOSPHOINOSITIDE 3-KINASE DELTA SYNDROME 1 (APDS1) TREATED WITH HEMATOPOIETIC STEM CELL TRANSPLANTATION (HSCT): A 39-MONTH FOLLOW-UP AND LITERATURE REVIEW

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Introduction

- Autosomal Dominant Primary combined immunodeficiency disorder
- **APDS** - **p**110δ-**A**ctivating mutation causing **S**enescent **T** cells, **L**ymphadenopathy, and **I**mmunodeficiency **(PASLI disease) - Activated PI3Kδ - senescent T cells**
  - Mutation PIK3CD gene –- activated catalytic subunit p110δ - **APDS1**
  - Mutation PIK3R1 gene – activated regulatory subunit of PI3Kδ - **APDS2**
- Clinical spectrum – lymphoproliferation and autoimmunity
- Vasculitis has been reported in 6 CASES OF APDS – **Digital** and **cutaneous** vasculitis
- CNS vasculitis **not reported so far**
- **Impact of HSCT on associated vasculitis not been thoroughly studied**

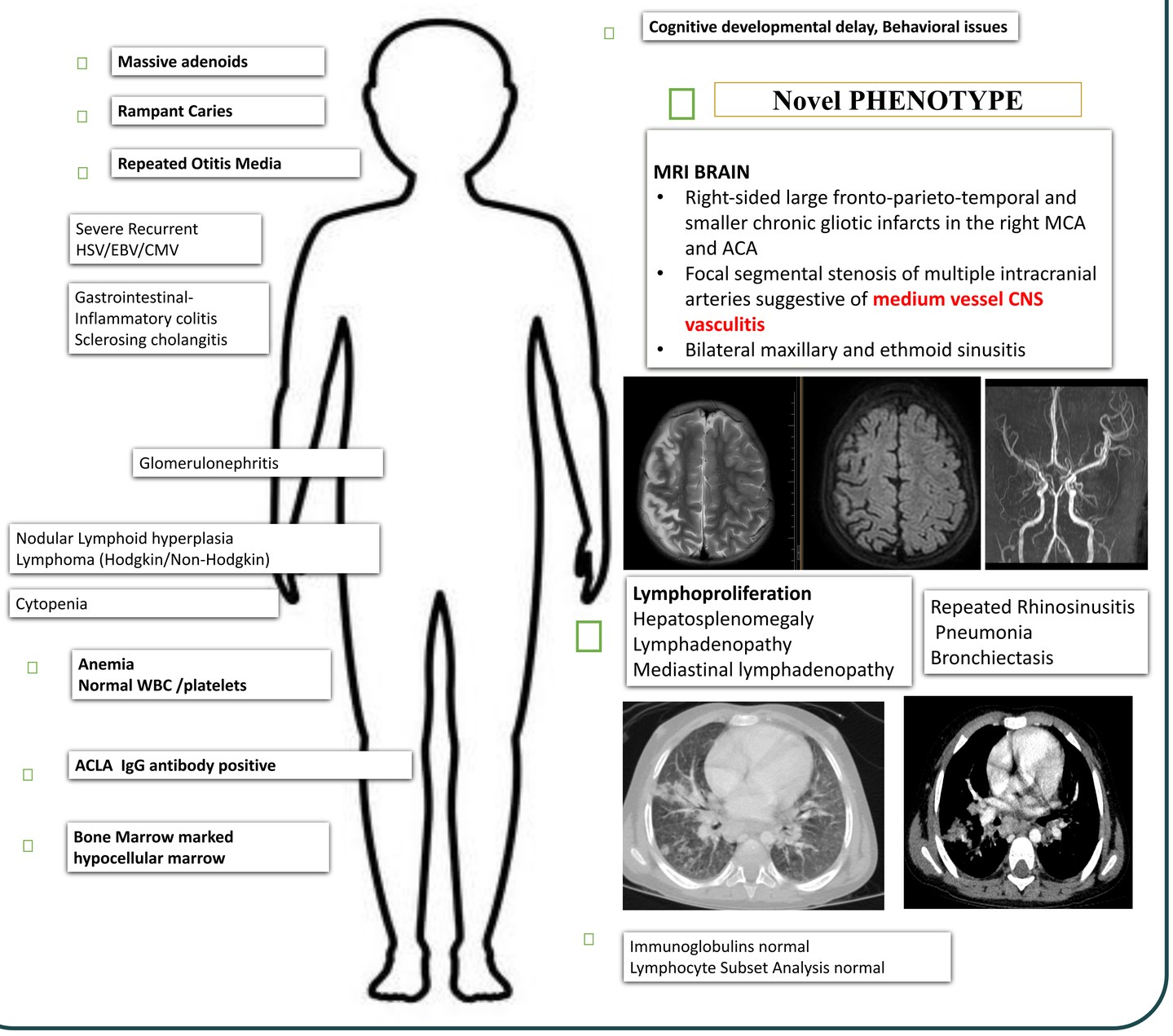
Clinical presentation

- Non-Consanguineous, F/ Full term normal delivery
- Birth weight: 3kg
- Uneventful neonatal period
- **acute onset weakness** of the left upper limb and upper motor neuron type left facial nerve palsy
- Weight 14 kg (3rd-50th percentile)
- Height: 91 cm (<3rd percentile)
- (Refer for other phenotypic features in cartoon)

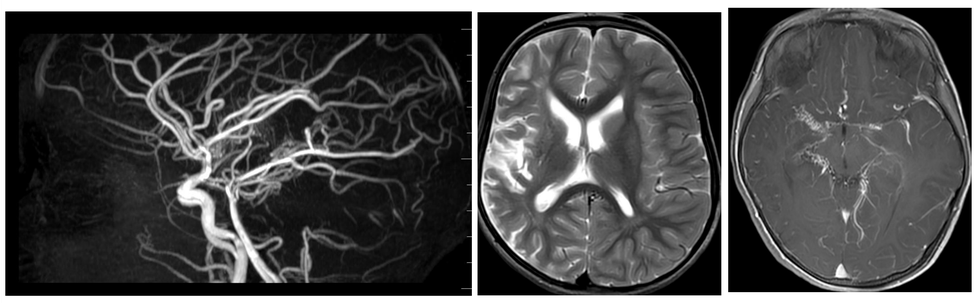
Treatment

- **Bone marrow transplant**
- Adenoidectomy
- Complete dental treatment

Phenotype in our case



Post HSCT MRI Angiography - 39 months



- Post Ischaemic changes in the MCA and ICA territory in the form of interval atrophy/ encephalomalacia in the right fronto-parieto-temporal and right basal ganglia.
- **No new areas of interval infarct**
- **Resolution of the previously seen areas of altered signal intensity in the left cerebral hemispheres.**
- Persistence of narrowing at the bilateral Supraglenoid ICA
- Complete occlusion of the right M1 segment of the MCA with **extensive collateral formations**
- **No new areas of vessel irregularity or occlusion or stenosis**

Post HSCT Phenotype- 39 months

- Absence of frequent infections.
- **Stabilization of CNS vasculitis** on serial MRIs, with marked vascular improvement and collateral formation
- Reversal of lymphoproliferation
- Normal immunological and cytokine profiles
- Significant physical growth **(weight gain: 9 kg,**75th percentile **and height increase: 32 cm** 75th percentile)
- Achievement of cognitive, speech, and developmental milestones comparable to peers.

Genotype

Gene & Transcript	Variant	Location	Zygosity	Disorder (OMIM)	Inheritance	Classification
PIK3CD NM 005026.5	c.3061G>A (p.Glu1021Lys)	Exon 24	Heterozygous	Immunodeficiency 14A (615513)	Autosomal Dominant	Pathogenic

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PO183

To Assess Health-Related Quality of Life (HRQoL) in Pediatric Patients with Hereditary Autoinflammatory Diseases (HAID) in India using KIDSCREEN - 52 Questionnaire - Pilot Study

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INTRODUCTION

- KIDSCREEN questionnaire - validated tool designed to evaluate HRQoL in children and adolescents
- 252 studies for chronic illnesses like cystic fibrosis, diabetes, Familial Mediterranean Fever, Juvenile Idiopathic Arthritis, mental health disorders et
- HAIDs - single study from Europe on 25 children reported
- KIDSCREEN 52 provides insights into specific areas of well-being affected by HAID.

AIM

- To evaluate **feasibility** and **assess HRQoL** in patients diagnosed with HAID
- Utilising parent-reported KIDSCREEN-52 data in their preferred language (English, Hindi)

METHOD

- Cross-sectional pilot study - from June 2024 to December 202
- KIDSCREEN-52 Parent Form, available in Hindi and English was administered to parents of children diagnosed with HAID.
- **Assesses ten domains** - :

physical well-being, psychological well-being, moods and emotions, autonomy, parent relations, social support, bullying, school environment, self-perception, and financial resources

Results

Domain	Average Rating	Standard deviation of ratings	Z-score	T-score
Activities	3.35	0.98	-1.90	30.99
Feelings	3.90	0.91	-0.15	48.49
Mood	4.18	0.90	0.75	57.49
About	4.25	0.77	0.98	59.77
Free Time	4.30	0.86	1.13	61.30
Family	4.19	0.86	0.84	58.44
Money	3.86	1.39	-0.28	47.22
Friends	3.55	1.16	-1.27	37.27
School	3.84	0.98	-0.33	46.72
Bullying	4.11	1.18	0.23	52.32

Descriptive statistics summarized findings. Standardised T-scores (normative average=50) and Z scores contextualised raw data, accounting for **discrepancies in number of questions in each domain**. Average ratings neutralize these differences, providing for a fairer assessment of HRQoL).

- T-scores **higher than 50 indicate better than average HRQoL**
- Z-scores measured **standard deviations from the mean, identifying outliers and variability across domains**.

Collection of data:

- Physical - (n=4)
- Virtually - Zoom platform (n=17).

Parents of 21 children (8 to 18 years) diagnosed with HAID answered the questionnaire.

Males: n=10
Females: n=11

Domains with Lowest T scores included:

- Physical Activities and Health (T-score = 30.99) reflecting significant limitations in physical functioning and mobility due to disease burden.

- Peer Relationships ("Friends") (T-score = 37.27) indicating challenges in social interactions.

Highest scores were observed in-

- "Free Time" (T-score = 61.30) indicating strong satisfaction with leisure opportunities.
- Emotional well-being, self-esteem, and family support: indicating resilience despite health challenges.

Variability:
Standard deviations ranged from 0.77–1.39, suggesting consistency in experiences across domains.
highest variability observed in the **'Money'** domain (SD = 1.39), reflecting diverse perceptions of financial circumstances.

Conclusion

Our study confirms the feasibility of KIDSCREEN-52 for assessing HRQoL in Indian pediatric patients and identifies the domains most affected in this pilot study.

References

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